Infimal-Convolution-Based Magnitude-Adaptive Regularization for Robust Graph Signal Recovery Against Graph Inaccuracies

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***Abstract*—Graph signal recovery commonly relies on graph Laplacian regularization (GLR) to promote smoothness over a given graph. In practice, however, the available graph is often only an imperfect proxy for the underlying graph-smoothness structure, and inaccuracies in its topology or edge weights can substantially distort the regularization. We characterize their effects through two failure modes: excessive penalties caused by spurious edges or overestimated weights, and information loss caused by missing edges or underestimated weights. Focusing on the former, we develop an infimal-convolution-based magnitude-adaptive regularization design whose resulting edge-domain penalty is a scaled separable Huber penalty applied to GLR-type proxy-weighted edge differences. The infimal-convolution representation provides a decomposition-based interpretation in which ordinary proxy-weighted edge differences are evaluated quadratically, while the threshold-exceeding portions of large differences are assigned to a linearly evaluated component. We further show that the resulting edge-domain penalty is the pointwise largest Lipschitz-continuous minorant of the quadratic GLR edge-domain penalty under a prescribed ℓ_1-Lipschitz constraint. We formulate graph signal recovery with this regularizer as a constrained convex optimization problem and derive an efficient solver based on preconditioned primal-dual splitting. Synthetic experiments under excessive-penalty and mixed proxy-graph inaccuracies show that the proposed method mitigates performance degradation when excessive penalties arise and retains its benefit when both types of failure modes coexist. Experiments on real-world meteorological signals further demonstrate its practical effectiveness when recovery relies on spatial proxy graphs that may naturally mismatch the underlying signal structure.**

***Index Terms*—Graph signal recovery, graph Laplacian regularization, infimal convolution, proximal splitting**

I. Introduction

Graph signal processing (GSP) provides a theoretical framework for analyzing and processing graph signals, i.e., signals whose values are associated with the nodes of a graph [1]–[3]. As the signal domain is generalized from regular Euclidean domains, such as time axes, image grids, and spatial coordinates, to graph-structured domains, GSP extends classical signal processing concepts, including Fourier analysis, frequency, filtering, sampling, and smoothness, to their graph-induced counterparts [4]–[6]. From a signal processing perspective, a graph can be viewed as a structural model that characterizes how a signal is expected to vary over its domain, with conventional regular grids included as special cases. This viewpoint enables flexible signal processing for signals naturally observed over networks as well as those arising in conventional data domains such as images, point clouds, and sensor data [7]–[12].

Graph signal recovery is a fundamental task in GSP, aiming to estimate an underlying graph signal from noisy, incomplete, or indirect observations [11], [13]–[15]. Since such observations generally provide insufficient information for stable recovery, a common approach is to formulate recovery as a regularized optimization problem, where the regularizer compensates for missing or degraded signal information by imposing prior knowledge on the target signal [16], [17]. In graph signal recovery, this prior knowledge has two complementary ingredients: a graph describing which signal samples are related and a signal model describing which variations over those relations are plausible. Accordingly, successful recovery requires both a graph that appropriately represents the relations among signal samples and a regularizer that appropriately expresses the target signal property over that graph.

Among various graph signal properties, graph smoothness provides a natural means of expressing prior knowledge through pairwise relations. Larger edge weights indicate a stronger expectation that the connected signal values should be similar, whereas smaller weights allow larger variations between them. For example, in an irregular sensor network, edge weights may encode spatial proximity or similarity in sensing conditions, so that graph smoothness expresses the expectation that strongly related sensors exhibit similar measurements without assuming a regular spatial grid. This representation has motivated many graph signal recovery methods based on smoothness-promoting regularization [18]–[23].

A representative smoothness-promoting regularizer is graph Laplacian regularization (GLR), which provides a standard means of quantifying and promoting graph smoothness [20], [24], [25]. For a weighted graph, GLR penalizes edge-wise signal differences according to their weights, or equivalently evaluates weight-modulated graph differences in a quadratic manner. When the given edge weights properly reflect the variation structure of the target signal, GLR provides a natural way to promote graph-smooth signals: differences across strongly connected nodes are strongly penalized, while those across weakly connected nodes are relatively tolerated. In this sense, GLR can be viewed as a graph-domain extension of classical quadratic smoothing, where the graph weights determine how signal differences should be balanced over the graph.

In practice, however, the graph available for signal recovery is often only an imperfect proxy for the underlying graph-smoothness structure, and graph-signal processing and recovery under mismatched or uncertain graph topologies have therefore been investigated in several studies [26]–[31]. In intrinsically networked domains, such as power grids, transportation systems, and water distribution networks, the topology or candidate edge weights may be obtained from physical connections and system parameters. Nevertheless, even a physically valid graph does not necessarily reflect

how the particular target signal is expected to vary, and its topology or weights may be affected by parameter uncertainty or changing operating conditions. In other applications, the graph is constructed from observed node features or pairwise relations; for example, its connectivity may be determined using a k-nearest-neighbor rule and its weights using a Gaussian kernel [32], [33]. Noise or missing information in these features can distort both the inferred neighborhood relations and the assigned similarities, potentially connecting samples across genuine spatial or semantic boundaries. Moreover, selecting suitable features, similarity measures, and graph-construction parameters is itself nontrivial. Consequently, the resulting proxy graph may contain spurious edges, missing edges, overestimated weights, or underestimated weights.

Such graph inaccuracies can substantially distort the smoothness evaluation induced by GLR. By changing which signal differences are penalized and how strongly they are weighted, an inaccurate graph can alter the relative regularization costs assigned to observation-consistent candidates. These inaccuracies give rise to two distinct failure modes. Spurious edges and overestimated weights can create excessively large proxy-weighted edge differences even for the underlying signal, thereby assigning it an undeservedly large regularization cost and biasing the recovery toward oversmoothed solutions. In contrast, missing edges and underestimated weights remove or weaken informative penalties, reducing the ability of GLR to distinguish the underlying signal from competing observation-consistent candidates. Both failure modes can substantially degrade recovery performance, but they require different remedies: the former calls for limiting excessive penalties introduced by the proxy graph, whereas the latter calls for recovering or supplementing missing graph information.

A natural graph-side response to this issue is to estimate a graph that more faithfully represents the variation structure of the signals of interest, so that GLR can provide an appropriate smoothness prior. Graph learning offers a principled framework for inferring graph structure from observed signals under assumptions such as signal smoothness [34]–[36]. For example, Dong et al. jointly estimate a graph Laplacian and smoothed representations of the input signals by minimizing their variations over the learned graph [35], while Berger et al. extend graph learning to noisy and incomplete observations through joint graph learning and signal inpainting [37]. Although these approaches reduce reliance on predefined graph structures or hand-crafted construction rules, the learned graph still depends on the available observations and on modeling assumptions imposed during graph inference [36], [37]. Consequently, a non-negligible mismatch may remain even when graph-learning techniques are employed.

Because graph inaccuracies cannot always be avoided in practice, a complementary regularizer-side approach is needed when signal recovery relies on an imperfect proxy graph. Rather than attempting to relearn or explicitly correct the graph, such an approach should make the regularization itself less sensitive to graph mismatch. In this work, we focus on inaccuracies that manifest as excessively large proxy-weighted edge differences and consequently assign undeservedly large regularization costs to the underlying signal. Our objective is not to identify inaccurate edges, reconstruct missing graph relations, or correct the proxy graph itself. Instead, we seek to limit excessive penalties while retaining the useful information contained in ordinary proxy-weighted edge differences. Such a design should not rely on prior knowledge of whether inaccuracies are present, what types they take, or which graph relations are affected, since such information is generally unavailable in practice. This naturally raises the following research question: *How can we limit the influence of excessive proxy-weighted edge differences while retaining the ability to promote the graph smoothness of the underlying signal, without prior knowledge of which graph relations are inaccurate?*

In this paper, we propose an infimal-convolution-based magnitude-adaptive regularizer for graph signal recovery to mitigate excessive penalties introduced by spurious edges and overestimated weights in a proxy graph. The proposed regularizer uses infimal convolution to decompose the proxy-weighted edge-difference vector into a quadratically evaluated component and a linearly evaluated component, allowing portions that would otherwise incur excessively large quadratic costs to be evaluated with slower growth. The penalty induced by this decomposition coincides with a scaled separable Huber penalty. Consequently, ordinary proxy-weighted edge differences are evaluated quadratically in a GLR-like manner, whereas the influence of excessively large differences is moderated through linear growth. This magnitude-adaptive evaluation requires no prior knowledge of whether graph inaccuracies are present, what types they take, or which graph relations are affected, while retaining the ability to promote the graph smoothness of the underlying signal.

The main contributions of this paper are summarized below.

- We develop an infimal-convolution-based magnitude-adaptive regularization design to mitigate excessive penalties caused by inaccuracies in a proxy graph. The resulting regularizer admits an equivalent scaled separable Huber representation, retaining GLR-like quadratic evaluation for ordinary proxy-weighted edge differences while limiting the influence of excessively large ones through linear growth. We further establish that, among edge-domain penalties satisfying a prescribed Lipschitz bound, the proposed penalty is the pointwise largest minorant of the quadratic GLR edge-domain penalty.
- We formulate graph signal recovery with the proposed regularizer as a general constrained convex optimization problem and establish the existence of optimal solutions under mild assumptions.
- We derive an efficient algorithm based on preconditioned primal–dual splitting, with computable proximal steps for the proposed regularization model.

Experiments are conducted on noisy graph signal inpainting tasks using both synthetic and real-world data. In the synthetic experiments, we first evaluate the proposed method under spurious-edge and overestimated-weight inaccuracies, which directly correspond to the excessive-penalty failure mode targeted in this work. We then consider a more realistic mixed-inaccuracy setting in which spurious edges, missing edges,

overestimated weights, and underestimated weights coexist, thereby introducing both excessive-penalty and information-loss effects. The proposed method is compared with GLR and potential alternatives, including GTV [14] and the sum of GLR and GTV. The results show that the proposed method mitigates performance degradation when graph inaccuracies induce excessive proxy-weighted edge differences and retains its benefit when multiple types of inaccuracies coexist. Experiments on real-world meteorological signals are further conducted to assess its practical effectiveness when recovery relies on naturally imperfect spatial proxy graphs.

II. Preliminaries

A. Notations and Mathematical Tools

Vectors are denoted by bold lowercase letters, with the i-th element of a vector $\mathbf{x}$ denoted by x_i. Matrices are denoted by bold uppercase letters, and the transpose of a matrix $\mathbf{X}$ is denoted by $\mathbf{X}^\top$. The set of all proper lower semicontinuous convex functions from $\mathbb{R}^m$ to $(-\infty, +\infty]$ is denoted by $\Gamma_0(\mathbb{R}^m)$. For a vector $\mathbf{x} \in \mathbb{R}^m$ and $p \geq 1$, the ℓ_p-norm is defined as $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$. For a matrix $\mathbf{A} \in \mathbb{R}^{n\times m}$, its operator norm is defined as $\|\mathbf{A}\|_{\mathrm{op}} = \sup_{\mathbf{x}\neq\mathbf{0}} \|\mathbf{A}\mathbf{x}\|_2/\|\mathbf{x}\|_2$. The indicator function $\iota_C \in \Gamma_0(\mathbb{R}^m)$ of a nonempty closed convex set $C \subset \mathbb{R}^m$ is defined as $\iota_C(\mathbf{x}) = 0$ if $\mathbf{x} \in C$ and $+\infty$ otherwise. For a function $f \in \Gamma_0(\mathbb{R}^m)$ and a parameter $\gamma > 0$, its proximity operator $\mathrm{prox}_{\gamma f} : \mathbb{R}^m \to \mathbb{R}^m$ is defined as

$$\mathrm{prox}_{\gamma f}(\mathbf{x}) = \underset{\mathbf{y}\in\mathbb{R}^m}{\mathrm{argmin}} \left\{ f(\mathbf{y}) + \frac{1}{2\gamma}\|\mathbf{y}-\mathbf{x}\|_2^2 \right\}. \tag{1}$$

We call a function proximable if its proximity operator has a closed-form expression or can be efficiently computed.

For two functions $f, g : \mathbb{R}^d \to (-\infty, +\infty]$, their infimal convolution [38] is defined as

$$(f \square g)(\mathbf{u}) = \inf_{\mathbf{u}=\mathbf{p}+\mathbf{q}} \{f(\mathbf{p}) + g(\mathbf{q})\}. \tag{2}$$

The infimal convolution determines the lowest total cost among all decompositions of $\mathbf{u}$ into two components evaluated by f and g, respectively.

B. Graph Signals and Graph Laplacian Regularization

We consider an undirected weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ without self-loops, where $\mathcal{V} = \{v_1, \ldots, v_{|\mathcal{V}|}\}$ is the set of vertices, $\mathcal{E} = \{e_{i,j} = (v_i, v_j)\}$ is the set of edges defined for some $i < j$, and $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the cardinalities of the corresponding sets. The diagonal matrix $\mathbf{W} \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{E}|}$ collects the positive edge weights $w_{i,j} > 0$ associated with $e_{i,j}$. The graph is commonly represented by its unnormalized graph Laplacian matrix $\mathbf{L} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$, which is defined from the weighted adjacency and degree matrices [24]. The graph Laplacian admits another equivalent expression as

$$\mathbf{L} = \mathbf{D}^\top \mathbf{W} \mathbf{D}, \tag{3}$$

where $\mathbf{D} \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{V}|}$ is an oriented incidence matrix defined by

$$D_{e_{i,j},v_k} := \begin{cases} -1, & \text{if } i = k, \\ 1, & \text{if } j = k, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

A graph signal is a vector $\mathbf{x} \in \mathbb{R}^{|\mathcal{V}|}$ whose entry x_i is associated with vertex v_i. A graph signal is regarded as smooth over $\mathcal{G}$ when its values tend to vary little between vertices that are strongly related by the graph. A standard measure of graph-signal smoothness is the quadratic form of the graph Laplacian, which gives the graph Laplacian regularization (GLR) [24], [25].

$$R_{\mathrm{GLR}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{L} \mathbf{x}. \tag{5}$$

A smaller value of $R_{\mathrm{GLR}}(\mathbf{x})$ indicates greater smoothness with respect to the graph structure encoded by $\mathbf{L}$, and GLR is widely used as a smoothness-promoting regularizer in graph signal recovery.

C. Preconditioned Primal-Dual Splitting (P-PDS)

P-PDS [39] can solve convex optimization problems of the following form:

$$\begin{aligned} \min_{\mathbf{x}_i, \mathbf{y}_j} \quad & \sum_{i=1}^{N} f_i(\mathbf{x}_i) + \sum_{j=1}^{M} g_j(\mathbf{y}_j) \\ \text{s.t.} \quad & \forall j \in \{1, \ldots, M\}, \quad \mathbf{y}_j = \sum_{i=1}^{N} \mathbf{L}_{j,i} \mathbf{x}_i, \end{aligned} \tag{6}$$

where $f_i \in \Gamma_0(\mathbb{R}^{n_i})$ and $g_j \in \Gamma_0(\mathbb{R}^{m_j})$ are proximable, and $\mathbf{L}_{j,i} \in \mathbb{R}^{m_j \times n_i}$ are arbitrary matrices. Let $\rho_i, \gamma_j > 0$ denote the stepsizes for the primal variable $\mathbf{x}_i$ and the dual variable $\mathbf{z}_j$, respectively. P-PDS performs the following updates for all i and j, with intermediate variables $\widetilde{\mathbf{z}}_j$:

$$\begin{aligned} \mathbf{x}_i^{(n+1)} &= \mathrm{prox}_{\rho_i f_i} \left(\mathbf{x}_i^{(n)} - \rho_i \sum_{j=1}^{M} \mathbf{L}_{j,i}^\top \mathbf{z}_j^{(n)} \right), \\ \tilde{\mathbf{z}}_j &= \mathbf{z}_j^{(n)} + \gamma_j \sum_{i=1}^{N} \mathbf{L}_{j,i} \left(2\mathbf{x}_i^{(n+1)} - \mathbf{x}_i^{(n)} \right), \\ \mathbf{z}_j^{(n+1)} &= \tilde{\mathbf{z}}_j - \gamma_j \, \mathrm{prox}_{\gamma_j^{-1} g_j} \left(\gamma_j^{-1} \tilde{\mathbf{z}}_j \right). \end{aligned} \tag{7}$$

Under the standard qualification condition for Problem (6) and appropriate stepsize conditions, the generated sequence converges to a primal-dual solution. A recent study [40] proposed three variable-wise stepsize designs based on upper bounds $\nu_{j,i}$ of the operator norms of the matrices $\mathbf{L}_{j,i}$ while guaranteeing the required stepsize conditions.

III. Proposed Regularizer

A. GLR under Graph Inaccuracies

We aim to design a regularizer that promotes graph smoothness using a potentially inaccurate proxy graph while retaining the useful information contained in the available graph structure. We neither refine the proxy graph itself nor assume prior knowledge of which graph relations are inaccurate, since such information is generally unavailable in practice. To identify the regularization properties required under such graph uncertainty, we first examine how graph inaccuracies affect GLR through its edge-domain expression.

Although GLR is typically written in the node domain as the quadratic form of the graph Laplacian, it can equivalently be expressed in the edge domain as

$$R_{\mathrm{GLR}}(\mathbf{u}) = \left\| \sqrt{\mathbf{W}}\mathbf{D}\mathbf{u} \right\|_2^2, \tag{8}$$

which follows from $\mathbf{L} = \mathbf{D}^\top \mathbf{W}\mathbf{D}$, as stated in Section II. Here, $\mathbf{D}$ and $\mathbf{W}$ represent the edge connectivity and edge weights of the available proxy graph, respectively. Accordingly, we refer to the entries of $\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}$ as proxy-weighted signal differences. Topology inaccuracies alter which signal differences are represented through $\mathbf{D}$, whereas edge-weight inaccuracies alter how strongly those differences are scaled through $\mathbf{W}$. Although these inaccuracies arise from different graph-side mechanisms, their effects on GLR can thus be interpreted through distortions of the proxy-weighted signal differences.

Accordingly, in the remainder of this section, we analyze the robustness of the regularizer to graph inaccuracies primarily through the proxy-weighted signal differences $\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}$, rather than directly through $\mathbf{D}$ and $\mathbf{W}$. Under the quadratic evaluation of GLR, these distortions can lead to two distinct failure modes:

- *Excessive-penalty failure mode:* spurious edges and overestimated weights can associate large edge weights with large signal differences, producing excessively large proxy-weighted signal differences. Their quadratic evaluation can then dominate the GLR cost and undesirably suppress legitimate signal variations.
- *Information-loss failure mode:* missing edges and underestimated weights can associate small or zero edge weights with informative signal differences, producing weakened or missing proxy-weighted signal differences. Consequently, competing observation-consistent signals can receive insufficient regularization and become difficult to distinguish from the underlying signal.

In this paper, we focus on the excessive-penalty failure mode and consider a magnitude-adaptive smoothness regularization design that mitigates the excessive influence of proxy-weighted signal differences without identifying or correcting inaccurate graph relations themselves. Although the proposed regularizer does not directly address information loss caused by missing edges or underestimated weights, we also consider mixed-inaccuracy settings in the experiments to assess whether its benefit is retained when both failure modes coexist.

B. Magnitude-Adaptive Regularization via Infimal Convolution

Building on the preceding analysis, our goal is to design a regularizer that preserves the GLR evaluation as much as possible while preventing excessively large proxy-weighted signal differences from dominating the quadratic evaluation. To this end, we use infimal convolution to decompose the proxy-weighted signal differences into two components, one evaluated quadratically as in GLR and the other redirected to an alternative penalty when it would otherwise incur an excessively large quadratic cost.

Definition 1 (Proposed regularizer)**.** *For $\tau > 0$, the proposed regularizer is defined as*

$$R_{\mathrm{IC}}(\mathbf{u}) = \Psi_\tau \left(\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u} \right), \tag{9}$$

where $\Psi_\tau : \mathbb{R}^{|\mathcal{E}|} \to \mathbb{R}$ is defined by

$$\Psi_\tau(\mathbf{z}) := \inf_{\mathbf{z}=\mathbf{p}+\mathbf{q}} \left\{ \|\mathbf{p}\|_2^2 + 2\tau \|\mathbf{q}\|_1 \right\}. \tag{10}$$

In this construction, $\mathbf{p}$ is assigned a quadratic cost as in GLR, whereas $\mathbf{q}$ is assigned a linear cost, whose growth is controlled by the parameter τ. Rather than evaluating $\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}$ entirely through the quadratic cost, the optimization over its decompositions can redirect portions that would otherwise incur excessive quadratic costs to $\mathbf{q}$, for which the assigned ℓ_1 cost grows more slowly than the quadratic cost evaluated at $\mathbf{p}$. This mechanism depends only on the magnitudes of the proxy-weighted signal differences and therefore requires no prior knowledge of the presence, types, or locations of graph inaccuracies.

The resulting edge-domain penalty admits a standard Huber characterization through its Moreau-envelope representation [41]. The following fact makes this relationship explicit and will also be used in the subsequent analysis.

Fact 2 (Moreau-envelope and Huber characterization)**.** *The scaled function $\Psi_\tau/(2\tau)$ is the Moreau envelope of the ℓ_1 norm with parameter τ. Moreover, Ψ_τ is separable and can be written as*

$$\Psi_\tau(\mathbf{z}) = 2 \sum_{e \in \mathcal{E}} H_\tau(z_e),$$

where

$$H_\tau(t) = \begin{cases} \frac{1}{2}t^2, & |t| \le \tau, \\ \tau|t| - \frac{1}{2}\tau^2, & |t| > \tau, \end{cases}$$

is the scalar Huber function with threshold τ.

Thus, the penalty induced by the infimal-convolution design is a scaled separable Huber penalty acting on the GLR-type proxy-weighted signal differences $\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}$. By the standard properties of Moreau envelopes, Ψ_τ is finite-valued, convex, and continuous, and so is R_{IC} by (9).

We next use the infimal-convolution representation to characterize how the proxy-weighted signal differences are optimally decomposed between the quadratic and linear components. This characterization makes the magnitude-adaptive allocation induced by the proposed regularizer explicit. For notational convenience, let

$$\mathbf{z} := \sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}. \tag{11}$$

Proposition 3 (Optimal decomposition)**.** *The infimum defining $R_{\mathrm{IC}}(\mathbf{u})$ is attained uniquely at*

$$\mathbf{q}^\star = \mathrm{soft}(\mathbf{z}; \tau), \qquad \mathbf{p}^\star = \mathrm{clip}(\mathbf{z}; \tau), \tag{12}$$

where the operators are defined componentwise as

$$[\mathrm{soft}(\mathbf{z}; \tau)]_e := \mathrm{sign}(z_e) \max\{|z_e| - \tau, 0\}, \tag{13}$$

$$[\mathrm{clip}(\mathbf{z}; \tau)]_e := \mathrm{sign}(z_e) \min\{|z_e|, \tau\}, \tag{14}$$

with $\mathrm{sign}(0) = 0$.

Proof: Eliminating $\mathbf{p}$ from (10) shows that the infimum is attained at $\mathbf{q}^\star = \mathrm{prox}_{\tau\|\cdot\|_1}(\mathbf{z})$, which is the well-known soft-thresholding operator given by (13). The equality constraint then gives $\mathbf{p}^\star = \mathbf{z} - \mathbf{q}^\star = \mathrm{clip}(\mathbf{z};\tau)$. ∎

Proposition 3 formalizes the intended magnitude-adaptive behavior governed by the threshold τ. If $|z_e| \le \tau$, then $q_e^\star = 0$, and the corresponding proxy-weighted signal difference remains entirely in the quadratically evaluated component. If $|z_e| > \tau$, then $|p_e^\star| = \tau$, and the threshold-exceeding portion $|z_e| - \tau$ is assigned to $q_e^\star$. The optimal decomposition further reveals how the proposed regularization value differs from the original GLR cost.

Proposition 4 (Relation to GLR)**.** *For any $\mathbf{u} \in \mathbb{R}^{|\mathcal{V}|}$, the proposed regularizer satisfies*

$$R_{\mathrm{IC}}(\mathbf{u}) = R_{\mathrm{GLR}}(\mathbf{u}) - \|\mathbf{q}^\star\|_2^2, \tag{15}$$

where $\mathbf{q}^\star$ is given in Proposition 3. Moreover,

$$R_{\mathrm{IC}}(\mathbf{u}) = R_{\mathrm{GLR}}(\mathbf{u}) \quad \Longleftrightarrow \quad \left\|\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}\right\|_\infty \le \tau. \tag{16}$$

Proof: Let $\mathbf{z} := \sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}$. Notice that the Huber function in Fact 2 can equivalently be written as

$$2H_\tau(z_e) = z_e^2 - \left(\max\{|z_e| - \tau, 0\}\right)^2.$$

By Proposition 3, $|q_e^\star| = \max\{|z_e| - \tau, 0\}$. Hence, summing the above relation over $e \in \mathcal{E}$ and using (8) gives (15). Equation (15) gives equality with GLR if and only if $\mathbf{q}^\star = \mathbf{0}$. By Proposition 3, this condition is equivalent to $\|\mathbf{z}\|_\infty \le \tau$, which, together with (11), proves (16). ∎

Proposition 4 shows that the proposed regularizer coincides with GLR when no proxy-weighted signal difference exceeds the threshold. When threshold-exceeding components are present, the proposed regularizer reduces the GLR cost by $\|\mathbf{q}^\star\|_2^2$, thereby suppressing their excessive contribution.

Remark 1 (Sensitivity to excessive edge weights)**.** *By Fact 2, the contribution associated with each proxy-weighted signal difference z_e grows quadratically for $|z_e| \le \tau$ and linearly as $2\tau|z_e| - \tau^2$ for $|z_e| > \tau$. As an illustrative example, consider an individual proxy edge e and fix its signal difference $d_e := (\mathbf{D}\mathbf{u})_e \neq 0$. Its proxy-weighted signal difference is $z_e = \sqrt{w_e}d_e$. Once $\sqrt{w_e}|d_e| > \tau$, the corresponding contribution to the proposed regularizer is*

$$2\tau\sqrt{w_e}|d_e| - \tau^2. \tag{17}$$

Hence, for fixed d_e, the contribution grows as $\Theta(\sqrt{w_e})$ with the proxy edge weight, whereas the corresponding GLR contribution $w_e d_e^2$ grows as $\Theta(w_e)$. This reduced growth characterizes the sensitivity of the proposed regularizer to an overestimated edge weight after the resulting proxy-weighted signal difference exceeds the threshold. The same argument applies to a spurious edge by interpreting $w_e > 0$ as the proxy weight assigned to the newly introduced edge.

Finally, we establish the smoothness of the proposed regularizer. This property characterizes its mathematical regularity and will also be useful for optimization.

Proposition 5 (Gradient and smoothness of the proposed regularizer)**.** *The proposed regularizer is differentiable with gradient*

$$\nabla R_{\mathrm{IC}}(\mathbf{u}) = 2\mathbf{D}^\top\sqrt{\mathbf{W}}\,\mathrm{clip}\left(\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u};\tau\right). \tag{18}$$

Moreover, ∇R_{IC} is $2\lambda_{\max}(\mathbf{L})$-Lipschitz continuous with respect to the ℓ_2 norm.

Proof: By the Moreau-envelope characterization in Fact 2 and the standard properties of the Moreau envelope [38, Prop. 12.30],

$$\nabla\Psi_\tau(\mathbf{z}) = 2\left(\mathbf{z} - \mathrm{prox}_{\tau\|\cdot\|_1}(\mathbf{z})\right), \tag{19}$$

and $\nabla\Psi_\tau$ is 2-Lipschitz continuous. Let $\mathbf{A} := \sqrt{\mathbf{W}}\mathbf{D}$. Applying the chain rule to (9) gives

$$\nabla R_{\mathrm{IC}}(\mathbf{u}) = \mathbf{A}^\top\nabla\Psi_\tau\left(\mathbf{A}\mathbf{u}\right), \tag{20}$$

which, together with (19) and Proposition 3, yields (18). For any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{|\mathcal{V}|}$, the 2-Lipschitz continuity of $\nabla\Psi_\tau$ gives

$$\begin{aligned}\|\nabla R_{\mathrm{IC}}(\mathbf{u}) - \nabla R_{\mathrm{IC}}(\mathbf{v})\|_2 &\le 2\left\|\mathbf{A}^\top\right\|_{\mathrm{op}}\|\mathbf{A}(\mathbf{u}-\mathbf{v})\|_2 \\ &\le 2\|\mathbf{A}\|_{\mathrm{op}}^2\|\mathbf{u}-\mathbf{v}\|_2 \\ &= 2\lambda_{\max}(\mathbf{L})\|\mathbf{u}-\mathbf{v}\|_2, \end{aligned} \tag{21}$$

where the third line follows from $\|\mathbf{A}\|_{\mathrm{op}}^2 = \lambda_{\max}(\mathbf{A}^\top\mathbf{A})$ and (3). ∎

Remark 2 (Proximability and splitting)**.** *Although $R_{\mathrm{IC}} = \Psi_\tau \circ (\sqrt{\mathbf{W}}\mathbf{D})$ is not proximable in general, Ψ_τ itself is proximable, with its proximity operator available in closed form as given later in* (36)*. Accordingly, optimization problems involving R_{IC} can be handled by suitable splitting algorithms. A specific instance of such an algorithm is presented in Section IV.*

C. Edge-Domain Sensitivity Analysis and Comparison with Potential Alternatives

In this subsection, we characterize the robustness of the proposed regularizer through the Lipschitz continuity of its associated edge-domain penalty Ψ_τ. In addition, we compare it with potential alternatives and establish an optimality property of the proposed edge-domain penalty as a replacement for the quadratic one in GLR under a prescribed Lipschitz constraint.

Proposition 6 (Lipschitz continuity and perturbation sensitivity)**.** *For all $\mathbf{z}, \widetilde{\mathbf{z}} \in \mathbb{R}^{|\mathcal{E}|}$, Ψ_τ is Lipschitz continuous with respect to ℓ_1 and ℓ_2 norms, specifically:*

$$|\Psi_\tau(\mathbf{z}) - \Psi_\tau(\widetilde{\mathbf{z}})| \le 2\tau\|\mathbf{z}-\widetilde{\mathbf{z}}\|_1 \le 2\tau\sqrt{|\mathcal{E}|}\|\mathbf{z}-\widetilde{\mathbf{z}}\|_2. \tag{22}$$

Consequently, for any perturbation $\boldsymbol{\delta} \in \mathbb{R}^{|\mathcal{E}|}$,

$$|\Psi_\tau(\mathbf{z}+\boldsymbol{\delta}) - \Psi_\tau(\mathbf{z})| \le 2\tau\|\boldsymbol{\delta}\|_1 \le 2\tau\sqrt{|\mathcal{E}|}\|\boldsymbol{\delta}\|_2. \tag{23}$$

Proof: For arbitrary $\mathbf{z}$ and $\widetilde{\mathbf{z}}$, the triangle inequality gives

$$\begin{aligned}\Psi_\tau(\mathbf{z}) &= \min_{\mathbf{y}}\left\{\|\mathbf{y}\|_2^2 + 2\tau\|\mathbf{z}-\mathbf{y}\|_1\right\} \\ &\le \min_{\mathbf{y}}\left\{\|\mathbf{y}\|_2^2 + 2\tau\|\widetilde{\mathbf{z}}-\mathbf{y}\|_1\right\} + 2\tau\|\mathbf{z}-\widetilde{\mathbf{z}}\|_1 \\ &= \Psi_\tau(\widetilde{\mathbf{z}}) + 2\tau\|\mathbf{z}-\widetilde{\mathbf{z}}\|_1, \end{aligned} \tag{24}$$

where we use $\min$ instead of $\inf$ since attainment was established in Proposition 3. Interchanging $\mathbf{z}$ and $\widetilde{\mathbf{z}}$ proves the first inequality in (22), while the norm equivalence $\|\mathbf{x}\|_1 \leq \sqrt{|\mathcal{E}|}\|\mathbf{x}\|_2$ gives the second. Equation (23) follows by setting $\widetilde{\mathbf{z}} = \mathbf{z} + \boldsymbol{\delta}$. ∎

Equation (23) shows that the change in the regularization value is uniformly bounded in terms of the perturbation magnitude, independently of the original proxy-weighted signal-difference vector. In contrast, the quadratic edge-domain penalty of GLR is not globally Lipschitz continuous and therefore does not admit such a uniform perturbation bound.

A natural alternative for obtaining bounded sensitivity is graph total variation (GTV) regularization, defined as

$$R_{\mathrm{GTV}}(\mathbf{u}) = \|\mathbf{W}\mathbf{D}\mathbf{u}\|_1 = \sum_{(i,j)\in\mathcal{E}} w_{i,j}|u_i - u_j|. \tag{25}$$

Let $\mathbf{r} := \mathbf{W}\mathbf{D}\mathbf{u}$ and $G(\mathbf{r}) := \|\mathbf{r}\|_1$ denote the weighted edge-difference vector and its edge-domain penalty, respectively. By the triangle inequality and norm equivalence,

$$|G(\mathbf{r}) - G(\widetilde{\mathbf{r}})| \leq \|\mathbf{r} - \widetilde{\mathbf{r}}\|_1 \leq \sqrt{|\mathcal{E}|}\|\mathbf{r} - \widetilde{\mathbf{r}}\|_2. \tag{26}$$

Thus, GTV has globally bounded sensitivity to perturbations of the weighted edge-difference vector and avoids the unbounded quadratic sensitivity of GLR to large weighted signal differences. However, its ℓ_1 evaluation promotes sparsity of the weighted edge differences, corresponding to a piecewise-smooth signal model rather than the quadratic graph smoothness promoted by GLR, which can lead to the well-known staircasing effect of TV-type regularization [42].

Since GTV provides bounded sensitivity but promotes a different signal model from GLR, another natural alternative is to supplement GLR with GTV, aiming to retain the quadratic graph-smoothness prior while incorporating the bounded-sensitivity property of the ℓ_1 term. We therefore consider

$$R_{\mathrm{GLR+GTV}}(\mathbf{u}) = \left\|\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}\right\|_2^2 + 2\tau\left\|\mathbf{W}\mathbf{D}\mathbf{u}\right\|_1. \tag{27}$$

However, since the quadratic GLR term remains unchanged, the resulting regularizer still has unbounded sensitivity to large proxy-weighted signal differences, and hence the excessive-penalty issue is not eliminated. This provides an illustrative example of the difficulty of achieving robustness through Lipschitz continuity while preserving the quadratic graph-smoothness prior of GLR.

As shown in Subsection III-B, the proposed regularizer retains the GLR evaluation for ordinary proxy-weighted signal differences while adaptively moderating the penalty growth for threshold-exceeding components, thereby achieving the global Lipschitz continuity established in Proposition 6. The following proposition shows that the edge-domain penalty Ψ_τ associated with the proposed regularizer preserves the original quadratic GLR evaluation as faithfully as possible while satisfying the prescribed Lipschitz constraint.

Proposition 7 (Largest Lipschitz minorant of GLR)**.** *The function Ψ_τ is the pointwise largest 2τ-Lipschitz continuous minorant of $Q(\mathbf{z}) = \|\mathbf{z}\|_2^2$ with respect to the ℓ_1 norm. That is, for any 2τ-Lipschitz continuous function $g : \mathbb{R}^{|\mathcal{E}|} \to \mathbb{R}$ satisfying $g(\mathbf{z}) \leq Q(\mathbf{z})$ for all $\mathbf{z} \in \mathbb{R}^{|\mathcal{E}|}$,*

$$g(\mathbf{z}) \leq \Psi_\tau(\mathbf{z}) \leq Q(\mathbf{z}), \qquad \forall\, \mathbf{z} \in \mathbb{R}^{|\mathcal{E}|}. \tag{28}$$

Proof: By Definition 1, choosing $\mathbf{p} = \mathbf{z}$ and $\mathbf{q} = \mathbf{0}$ gives $\Psi_\tau(\mathbf{z}) \leq Q(\mathbf{z})$. Moreover, Proposition 6 shows that Ψ_τ is 2τ-Lipschitz continuous with respect to the ℓ_1 norm. For any admissible g and any $\mathbf{y} \in \mathbb{R}^{|\mathcal{E}|}$, the Lipschitz property and the minorant condition give

$$\begin{aligned} g(\mathbf{z}) &\leq g(\mathbf{y}) + 2\tau\|\mathbf{z} - \mathbf{y}\|_1 \\ &\leq Q(\mathbf{y}) + 2\tau\|\mathbf{z} - \mathbf{y}\|_1. \end{aligned} \tag{29}$$

Taking the minimum over $\mathbf{y}$ yields

$$g(\mathbf{z}) \leq \min_{\mathbf{y}} \{Q(\mathbf{y}) + 2\tau\|\mathbf{z} - \mathbf{y}\|_1\} = \Psi_\tau(\mathbf{z}), \tag{30}$$

where the equality follows from Definition 1 by setting $\mathbf{p} = \mathbf{y}$ and $\mathbf{q} = \mathbf{z} - \mathbf{y}$. This proves (28). ∎

In the present graph-regularization setting, this result provides a precise design interpretation of the Huber form in Fact 2: under the prescribed Lipschitz constraint, Ψ_τ preserves the quadratic GLR edge-domain penalty pointwise as much as possible. We include GTV and GLR+GTV in the experiments to evaluate how these different regularization mechanisms affect recovery under proxy-graph inaccuracies.

IV. Problem Formulation and Optimization Algorithm

A. Optimization Problem

We formulate a broad class of smooth graph signal recovery problems as a unified constrained convex optimization problem incorporating the proposed regularizer to mitigate excessive penalties induced by an inaccurate proxy graph.

Let $\mathbf{u}^* \in \mathbb{R}^{|\mathcal{V}|}$ be the smooth graph signal to be recovered. We consider the following linear observation model:

$$\mathbf{v} = \boldsymbol{\Phi}\mathbf{u}^* + \mathbf{n}, \tag{31}$$

where $\mathbf{v} \in \mathbb{R}^K$ is the observed signal, and $\mathbf{n} \in \mathbb{R}^K$ is additive Gaussian noise. The linear observation matrix $\boldsymbol{\Phi} \in \mathbb{R}^{K\times|\mathcal{V}|}$ specifies the graph signal recovery task of interest, e.g., denoising when $\Phi = I$, inpainting when Φ is a binary masking matrix, and compressed sensing when Φ is a general sensing matrix.

Under this observation model, we formulate the recovery of the smooth graph signal $\mathbf{u}^*$ as the following constrained convex optimization problem using our proposed regularizer:

$$\min_{\mathbf{u}} \quad R_{\mathrm{IC}}(\mathbf{u}) \quad \text{s.t.} \quad \begin{cases} \boldsymbol{\Phi}\mathbf{u} \in B_{\mathbf{v},r}, \\ \mathbf{u} \in [\underline{\mu}, \overline{\mu}]^{|\mathcal{V}|}, \end{cases} \tag{32}$$

where $\mathbf{u}$ denotes the optimization variable corresponding to the recovered signal, and the closed convex constraint sets are defined as

$$B_{\mathbf{v},r} := \{\mathbf{z} \in \mathbb{R}^K \mid \|\mathbf{z} - \mathbf{v}\|_2 \leq r\}, \tag{33}$$

with the hyperparameters $r > 0$, and $\underline{\mu}, \overline{\mu} \in \mathbb{R}$.

The data-fidelity constraint $\boldsymbol{\Phi}\mathbf{u} \in B_{\mathbf{v},r}$ enforces the recovered signal and the observed signal to stay within a specified

distance in the observation domain, i.e., $\|\mathbf{\Phi u} - \mathbf{v}\|_2 \leq r$. Unlike a typical penalty formulation such as $\frac{\lambda}{2}\|\mathbf{\Phi u} - \mathbf{v}\|_2^2$ with a balancing parameter λ, this constrained formulation directly specifies the admissible discrepancy between $\mathbf{\Phi u}$ and $\mathbf{v}$ by r, not in a relative balance with other terms. This independence allows a direct interpretation of data fidelity and facilitates principled hyperparameter control [43]–[47]. In particular, r can be determined from the standard deviation σ of the Gaussian noise $\mathbf{n}$ as $r = \theta\sigma\sqrt{K}$, where $\theta > 0$ is a practical adjustment parameter, typically chosen to be close to 1.

The box constraint $\mathbf{u} \in [\underline{\mu}, \overline{\mu}]^{|\mathcal{V}|}$ enforces the recovered signal to stay within the specified value range, i.e., $\underline{\mu} \leq u_i \leq \overline{\mu}$. The upper and lower bounds, $\underline{\mu}$ and $\overline{\mu}$, can be set to the known range of signal values or sufficiently large when such prior knowledge is unavailable. Additionally, the box constraint ensures Problem (32) is well-posed in the sense that it admits at least one optimal solution, provided that the feasible set is nonempty. Specifically, this constraint makes the feasible set $\mathcal{S} := [\underline{\mu}, \overline{\mu}]^{|\mathcal{V}|} \cap \mathbf{\Phi}^{-1}(B_{\mathbf{v},r})$ compact (closed and bounded), as $\mathcal{S}$ is a closed subset of the compact set $[\underline{\mu}, \overline{\mu}]^{|\mathcal{V}|}$, which follows from the fact that $\mathbf{\Phi}^{-1}(B_{\mathbf{v},r}) = \{\mathbf{z} \mid \mathbf{\Phi z} \in B_{\mathbf{v},r}\}$ is closed. Thus, the lower semicontinuous objective function R_{IC} attains its infimum over the feasible set.

B. P-PDS-Based Optimization Algorithm

We develop an efficient algorithm for solving Problem (32) based on P-PDS, which requires neither matrix inversions nor inner iterative solvers and allows convergence-guaranteed variable-wise stepsize selection. Using $R_{\mathrm{IC}}(\mathbf{u}) = \Psi_\tau(\sqrt{\mathbf{W}}\mathbf{D}\mathbf{u})$ and the indicator functions of the constraint sets, Problem (32) can be equivalently reformulated as

$$\begin{aligned} \min_{\mathbf{u},\mathbf{y}_1,\mathbf{y}_2} \quad & \iota_{B_{\mathbf{v},r}}(\mathbf{y}_1) + \iota_{[\underline{\mu},\overline{\mu}]^{|\mathcal{V}|}}(\mathbf{u}) + \Psi_\tau(\mathbf{y}_2) \\ \text{s.t.} \quad & \begin{cases} \mathbf{y}_1 = \mathbf{\Phi u}, \\ \mathbf{y}_2 = \sqrt{\mathbf{W}}\mathbf{D}\mathbf{u}. \end{cases} \end{aligned} \tag{34}$$

This formulation fits the canonical P-PDS problem in (6) by setting $\mathbf{x}_1 = \mathbf{u}$, $f_1 = \iota_{[\underline{\mu},\overline{\mu}]^{|\mathcal{V}|}}$, $g_1 = \iota_{B_{\mathbf{v},r}}$, and $g_2 = \Psi_\tau$, with $\mathbf{L}_{1,1} = \mathbf{\Phi}$ and $\mathbf{L}_{2,1} = \sqrt{\mathbf{W}}\mathbf{D}$.

The required proximity operators are available in closed form. For $\kappa > 0$, the proximity operators of the two indicator functions are given by

$$\begin{aligned} &\mathrm{prox}_{\kappa\iota_{[\underline{\mu},\overline{\mu}]^{|\mathcal{V}|}}}(\mathbf{x}) = \max\big(\underline{\mu}, \min(\overline{\mu}, \mathbf{x})\big), \\ &\mathrm{prox}_{\kappa\iota_{B_{\mathbf{v},r}}}(\mathbf{x}) = \begin{cases} \mathbf{x}, & \|\mathbf{x} - \mathbf{v}\|_2 \leq r, \\ \mathbf{v} + \dfrac{r(\mathbf{x} - \mathbf{v})}{\|\mathbf{x} - \mathbf{v}\|_2}, & \text{otherwise}, \end{cases} \end{aligned} \tag{35}$$

where the maximum and minimum are taken componentwise. Moreover, as noted in Remark 2, Ψ_τ is proximable, and its proximity operator is given by

$$\mathrm{prox}_{\kappa\Psi_\tau}(\mathbf{x}) = \frac{1}{1 + 2\kappa}\left(\mathbf{x} + 2\kappa\,\mathrm{soft}\left(\mathbf{x}; (1 + 2\kappa)\tau\right)\right). \tag{36}$$

Hence, all proximity operations required by P-PDS can be evaluated explicitly. The resulting algorithm is summarized in Algorithm 1.

Algorithm 1 P-PDS-based algorithm for solving Problem (32)

Input: $\mathbf{u}^{(0)}$, $\mathbf{z}_1^{(0)}$, $\mathbf{z}_2^{(0)}$
$n \leftarrow 0$
while Stopping criteria are not met **do**
  $\widetilde{\mathbf{u}} \leftarrow \mathbf{\Phi}^\top \mathbf{z}_1^{(n)} + (\sqrt{\mathbf{W}}\mathbf{D})^\top \mathbf{z}_2^{(n)}$
  $\mathbf{u}^{(n+1)} \leftarrow \mathrm{prox}_{\rho_1 \iota_{[\underline{\mu},\overline{\mu}]^{|\mathcal{V}|}}}\left(\mathbf{u}^{(n)} - \rho_1 \widetilde{\mathbf{u}}\right)$
  $\widetilde{\mathbf{z}}_1 \leftarrow \mathbf{z}_1^{(n)} + \gamma_1 \mathbf{\Phi}\left(2\mathbf{u}^{(n+1)} - \mathbf{u}^{(n)}\right)$
  $\mathbf{z}_1^{(n+1)} \leftarrow \widetilde{\mathbf{z}}_1 - \gamma_1 \mathrm{prox}_{\gamma_1^{-1}\iota_{B_{\mathbf{v},r}}}\left(\gamma_1^{-1}\widetilde{\mathbf{z}}_1\right)$
  $\widetilde{\mathbf{z}}_2 \leftarrow \mathbf{z}_2^{(n)} + \gamma_2\sqrt{\mathbf{W}}\mathbf{D}\left(2\mathbf{u}^{(n+1)} - \mathbf{u}^{(n)}\right)$
  $\mathbf{z}_2^{(n+1)} \leftarrow \widetilde{\mathbf{z}}_2 - \gamma_2 \mathrm{prox}_{\gamma_2^{-1}\Psi_\tau}\left(\gamma_2^{-1}\widetilde{\mathbf{z}}_2\right)$
  $n \leftarrow n + 1$
end while
Output: $\mathbf{u}^{(n)}$

To determine the stepsizes, we employ the second variable-wise stepsize design proposed in [40], referred to therein as OVDP2. For the matrices in (34), let $\nu_{j,1}$ denote upper bounds satisfying $\|\mathbf{L}_{j,1}\|_{\mathrm{op}} \leq \nu_{j,1}$. For $\mathbf{L}_{1,1} = \mathbf{\Phi}$, any upper bound $\nu_{1,1} \geq \|\mathbf{\Phi}\|_{\mathrm{op}}$ can be used, and the exact operator norm can be obtained as the largest singular value of $\mathbf{\Phi}$. In particular, $\nu_{1,1} = 1$ can be used when $\mathbf{\Phi} = \mathbf{I}$ or when $\mathbf{\Phi}$ is a binary masking matrix. For $\mathbf{L}_{2,1} = \sqrt{\mathbf{W}}\mathbf{D}$, we take the exact operator norm $\nu_{2,1} = \|\sqrt{\mathbf{W}}\mathbf{D}\|_{\mathrm{op}} = \sqrt{\lambda_{\max}(\mathbf{L})}$, as shown in the proof of Proposition 5. Applying OVDP2 to these bounds yields

$$\begin{aligned} \rho_1 &= \frac{1}{\nu_{1,1} + \sqrt{\lambda_{\max}(\mathbf{L})}}, \\ \gamma_1 &= \frac{1}{\nu_{1,1}}, \qquad \gamma_2 = \frac{1}{\sqrt{\lambda_{\max}(\mathbf{L})}}. \end{aligned} \tag{37}$$

Under the standard qualification condition stated in Section II-C, the resulting sequence converges to a solution of Problem (32).

V. Experiments

We evaluate the proposed method on noisy graph signal inpainting tasks using both synthetic and real-world data. The proposed method is compared with graph Laplacian regularization (GLR), graph total variation (GTV), and the sum of GLR and GTV (GLR+GTV). In the synthetic experiments, we first examine recovery performance under spurious-edge and overestimated-weight inaccuracies, which directly induce the excessive-penalty failure mode targeted by the proposed regularizer. We then consider a mixed-inaccuracy setting in which excessive-penalty and information-loss effects coexist. Finally, we conduct experiments on real-world meteorological signals to assess the recovery performance when spatially constructed proxy graphs may naturally mismatch the underlying signal structure.

A. Common Experimental Protocol

We consider noisy graph signal inpainting under the observation model in (31). Unless otherwise noted, the underlying signal $\mathbf{u}^*$ has $|\mathcal{V}| = 256$ entries and is normalized to the range $[0, 1]$. The observation matrix $\mathbf{\Phi}$ is a binary masking matrix

that randomly masks 40% of the signal entries, and the additive Gaussian noise has standard deviation $\sigma = 0.05$.

For all experiments, the edge weights of each graph used for recovery are rescaled so that their mean is equal to one before applying the compared methods. This common normalization suppresses differences attributable merely to the global weight scale and allows the comparison to focus on the relative graph relations represented by the edge weights. It also provides a common scale for the proxy-weighted edge differences whose magnitudes are controlled by the threshold parameter τ in the proposed regularizer.

We evaluate GLR, GTV, GLR+GTV, and the proposed method within the constrained recovery framework introduced in Section IV-A, using identical data-fidelity and box constraints for all methods. The radius r of the data-fidelity ball $\mathcal{B}_{\mathbf{v},r}$ is set according to $r = \theta\sigma\sqrt{K}$, where K is the number of observed signal entries, and the adjustment parameter is set to $\theta = 0.9$. The box constraint is set to $[0,1]^{|\mathcal{V}|}$ because the underlying signals are normalized to $[0,1]$.

All recovery problems are solved using P-PDS with the same numerical accuracy settings. For each formulation, the stepsizes are determined using the variable-wise rule described in Section II-C, with exact operator norms used for the corresponding linear operators. Since $\mathbf{\Phi}$ is a binary masking matrix, its operator norm is one. All optimization variables are initialized to zero, and the iterations are terminated when the norm of the update between consecutive iterates, measured in the inner-product space associated with P-PDS [48], becomes smaller than 10^{-9} or when the number of iterations reaches 10^5. Recovery performance is evaluated using the normalized root mean squared error (NRMSE):

$$\text{NRMSE} = \frac{\|\widehat{\mathbf{u}} - \mathbf{u}^*\|_2}{\|\mathbf{u}^*\|_2}. \tag{38}$$

Here, $\widehat{\mathbf{u}}$ and $\mathbf{u}^*$ denote the recovered and underlying signals, respectively, and a smaller NRMSE indicates more accurate recovery. For each experimental condition, the reported performance is averaged over 20 independent trials with independently generated observation masks and noise realizations. For a given condition, the underlying signal and proxy graph are kept identical across all trials and methods.

For GLR+GTV and the proposed method, the parameter τ is selected from nine logarithmically spaced values ranging from 5×10^{-3} to 50. For each method, we select a single common value of τ for each experiment by minimizing the NRMSE aggregated over all evaluation conditions considered in that experiment. This selection is oracle because the NRMSE is evaluated using the underlying signals, but τ is not optimized separately for individual proxy-graph conditions. The conditions included in the aggregation are described in the corresponding experimental subsections. In addition, the synthetic experiments examine the sensitivity of the recovery performance to τ, showing similar trends across the different inaccuracy settings.

B. Experiments on Synthetic Data

1) Experimental Settings: To evaluate the proposed regularizer under controlled proxy-graph inaccuracies, we consider three types of synthetic inaccuracies: spurious edges, overestimated weights, and mixed inaccuracies. The first two isolate the excessive-penalty failure mode targeted by the proposed regularizer, whereas the mixed setting provides a scenario in which excessive-penalty and information-loss effects coexist.

TABLE I
SETTINGS FOR GENERATING PROXY-GRAPH INACCURACIES IN THE SYNTHETIC EXPERIMENTS. THE PARAMETERS ρ AND β DENOTE THE AFFECTED-EDGE FRACTION AND WEIGHT-MODIFICATION FACTOR, RESPECTIVELY.

Inaccuracy	Fraction ρ	Factor β
Spurious edges	0.05, 0.10, 0.15, 0.20	2.0
Overestimated weights	0.10	1.0, 2.0, 3.0, 4.0
Mixed inaccuracies	0.02, 0.04, 0.06, 0.08	2.0

To examine whether the observed behavior depends strongly on a particular underlying graph construction, we evaluate three graph-construction settings, denoted Settings I–III. In each setting, the underlying graph is a random geometric graph [49] with 256 nodes whose locations $\{\mathbf{p}_i\}_{i=1}^{256}$ are sampled from the uniform distribution over the unit square $[0,1]^2$. The graph topology is determined by a connection radius, and the edge weights are assigned using a Gaussian kernel. Settings I–III use the pairs $(0.14, 1.0)$, $(0.16, 1.0)$, and $(0.14, 0.4)$, respectively, where the first and second entries denote the connection radius and Gaussian-kernel standard deviation.

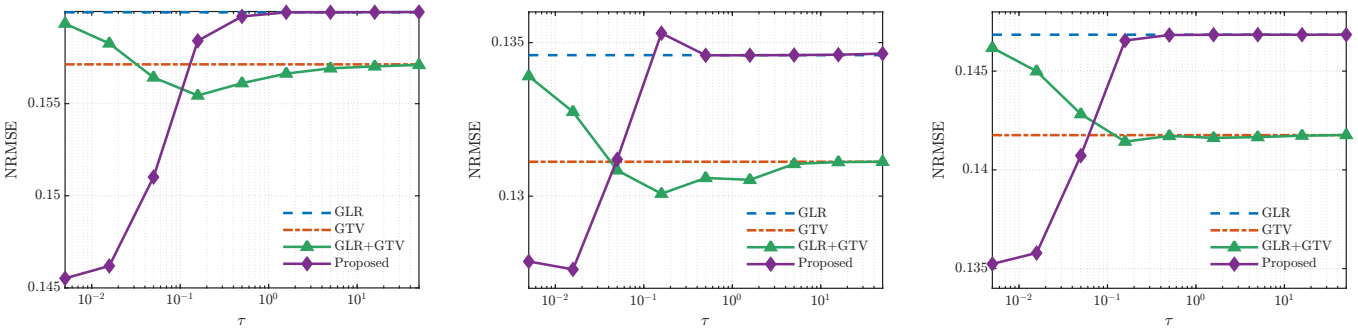
For each graph-construction setting, the underlying graph signal is generated as a random linear combination of the lowest-frequency 5% of the graph Fourier components of the corresponding underlying graph. The coefficients are independently drawn from a zero-mean Gaussian distribution with standard deviation one, and the resulting signal is normalized to the range $[0,1]$. Proxy graphs are then generated by perturbing the topology and edge weights of the corresponding underlying graph. The unmodified underlying graph is additionally evaluated as the no-inaccuracy (NI) reference.

The four types of proxy-graph inaccuracies used in the synthetic experiments are generated according to the following rules, where their severity is characterized by the affected-edge fraction ρ and the weight-modification factor β. Let $\mathcal{E}^*$ be the edge set of the underlying graph and let $w_{\max}$ and $w_{\min}$ denote its maximum and minimum edge weights, respectively. Spurious-edge inaccuracies are generated by randomly selecting $\rho|\mathcal{E}^*|$ node pairs that are not connected in the underlying graph and assigning each inserted edge the weight $\beta w_{\max}$. Missing-edge inaccuracies are generated by removing $\rho|\mathcal{E}^*|$ existing edges in descending order of their underlying weights. Overestimated-weight inaccuracies are generated by replacing the smallest selected $\rho|\mathcal{E}^*|$ edge weights with $\beta w_{\max}$, whereas underestimated-weight inaccuracies are generated by replacing the largest selected $\rho|\mathcal{E}^*|$ edge weights with $w_{\min}/\beta$. For the mixed-inaccuracy setting, the spurious- and missing-edge inaccuracies are introduced first, followed by the overestimated- and underestimated-weight inaccuracies on distinct remaining edges to avoid overlap among the affected edges. The same affected-edge fraction ρ is applied separately to all four inaccuracy types in the mixed setting. The parameter values

TABLE II

AVERAGE NRMSES UNDER SYNTHETIC PROXY-GRAPH INACCURACIES FOR THREE GRAPH-CONSTRUCTION SETTINGS. "NI" DENOTES THE NO-INACCURACY CONDITION. THE BEST AND SECOND-BEST VALUES WITHIN EACH GRAPH SETTING AND INACCURACY LEVEL ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

	Spurious edges (ρ)					**Overestimated weights** (β)					**Mixed inaccuracies** (ρ)				
Method	NI	0.05	0.10	0.15	0.20	NI	1	2	3	4	NI	0.02	0.04	0.06	0.08
Setting I (reference): (0.14, 1.0)															
GLR	0.1290	0.1481	0.1631	0.1760	0.1836	0.1290	0.1311	0.1351	0.1379	0.1399	0.1290	0.1342	0.1484	0.1571	0.1656
GTV	0.1216	0.1376	0.1581	0.1775	0.1909	0.1216	0.1253	0.1318	0.1366	0.1404	0.1216	**0.1259**	0.1392	0.1534	0.1687
GLR+GTV	**0.1208**	0.1405	0.1585	0.1737	0.1836	**0.1208**	**0.1240**	0.1305	0.1356	0.1394	**0.1208**	0.1265	0.1416	0.1538	0.1644
Proposed	0.1237	**0.1318**	**0.1452**	**0.1572**	**0.1697**	0.1233	0.1252	**0.1282**	**0.1299**	**0.1316**	0.1237	0.1265	**0.1337**	**0.1413**	**0.1510**
Setting II: (0.16, 1.0)															
GLR	0.1001	0.1212	0.1361	0.1422	0.1495	0.1001	0.1021	0.1063	0.1097	0.1125	0.1001	0.1097	0.1232	0.1320	0.1396
GTV	0.1019	0.1202	0.1377	0.1508	0.1615	0.1019	0.1041	0.1084	0.1143	0.1179	0.1019	0.1083	0.1225	0.1343	0.1470
GLR+GTV	**0.0991**	0.1199	0.1350	0.1419	0.1497	**0.0991**	0.1012	0.1059	0.1099	0.1132	**0.0986**	0.1082	0.1219	0.1317	0.1404
Proposed	0.1010	**0.1137**	**0.1261**	**0.1343**	**0.1438**	0.0996	**0.1010**	**0.1040**	**0.1062**	**0.1081**	0.1010	**0.1068**	**0.1153**	**0.1250**	**0.1324**
Setting III: (0.14, 0.4)															
GLR	0.0721	0.1227	0.1524	0.1694	0.1773	0.0721	0.0917	0.1007	0.1062	0.1103	0.0721	0.0971	0.1194	0.1422	0.1573
GTV	0.0660	0.1109	0.1497	0.1808	0.1863	0.0660	0.0831	0.0957	0.1097	0.1200	0.0660	0.0882	0.1100	0.1442	0.1665
GLR+GTV	**0.0620**	0.1175	0.1490	0.1698	0.1783	**0.0620**	0.0834	0.0969	0.1054	0.1117	**0.0619**	0.0881	0.1112	0.1423	0.1605
Proposed	0.0705	**0.0934**	**0.1126**	**0.1305**	**0.1451**	0.0705	**0.0816**	**0.0852**	**0.0888**	**0.0907**	0.0705	**0.0779**	**0.0949**	**0.1084**	**0.1176**



(a) Spurious edges (b) Overestimated weights (c) Mixed inaccuracies

Fig. 1. Average NRMSE versus the parameter τ for the reference graph setting under (a) spurious-edge, (b) overestimated-weight, and (c) mixed inaccuracies. Each value is averaged over all tested inaccuracy levels including the no-inaccuracy condition.

evaluated in the three inaccuracy experiments are summarized in Table I.

Following the common parameter-selection protocol in Section V-A, the reported results for each synthetic inaccuracy experiment correspond to a single common value of τ selected over all tested inaccuracy levels, rather than to values separately optimized for individual levels.

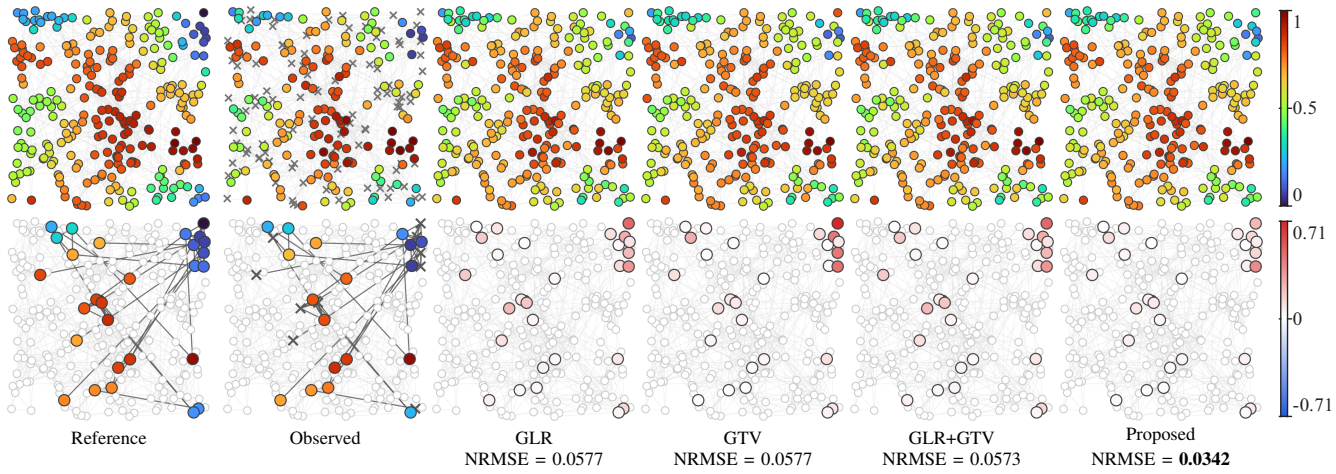
2) Results: Table II summarizes the average NRMSEs under the three types of proxy-graph inaccuracies for the three graph-construction settings. We first examine the spurious-edge and overestimated-weight experiments, which directly correspond to the excessive-penalty failure mode discussed in Section III-A.

For spurious-edge inaccuracies, the recovery performance of all methods generally degrades as the affected-edge fraction ρ increases. However, the NRMSE of the proposed method increases more gradually than those of the comparison methods across all three graph-construction settings. Although GTV or GLR+GTV can achieve lower NRMSE under the NI condition, the proposed method achieves the lowest NRMSE at every nonzero tested spurious-edge fraction in all three graph-construction settings.

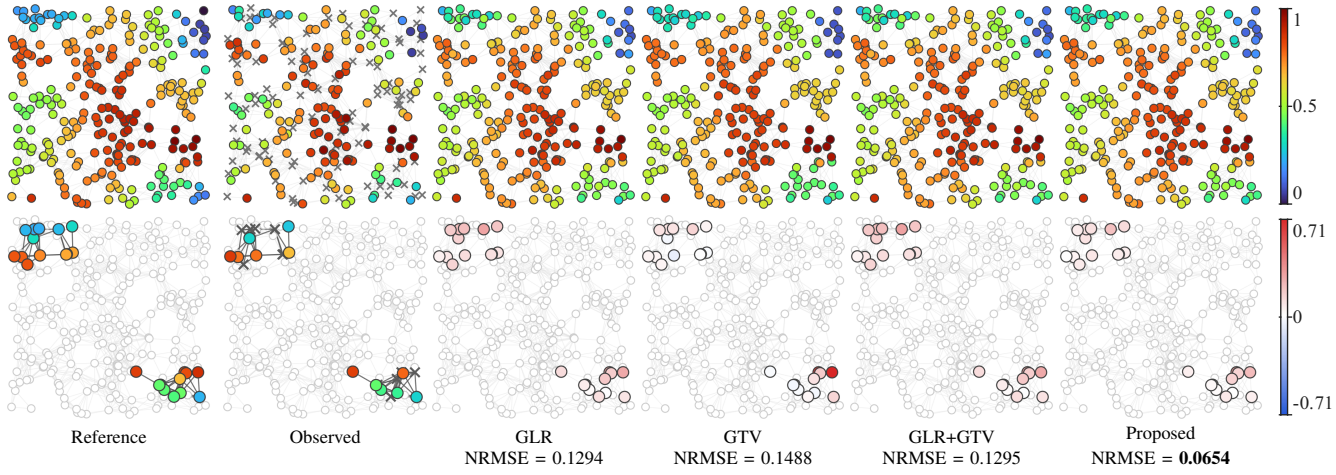
A similar trend is observed for overestimated weights. As β increases, thereby strengthening the weight overestimation while keeping the affected-edge fraction fixed, the relative advantage of the proposed method becomes more pronounced. In particular, the proposed method achieves the lowest NRMSE at the larger tested values of β in all three graph-construction settings.

The common behavior of these two experiments is important because the inaccuracies modify different aspects of the proxy graph. Spurious edges alter the graph topology, whereas overestimated weights preserve the topology and modify the strengths of existing graph relations. The observed trend is therefore consistent with the mechanism developed in Section III-B: the proposed regularizer limits the influence of excessively large proxy-weighted edge differences according to their magnitudes, rather than relying on a particular type of graph perturbation.

We next examine the mixed-inaccuracy experiment, in which excessive-penalty and information-loss effects coexist. As the common affected-edge fraction ρ increases, the



(a) Spurious edges



(b) Overestimated weights

Fig. 2. Qualitative recovery results with a connection radius of 0.14 and a Gaussian-kernel standard deviation of 0.4 under (a) spurious-edge inaccuracies with $\beta = 2$ and (b) overestimated-weight inaccuracies with $\rho = 0.1$. Upper rows show the reference, observed, and recovered signals; lower rows focus on nodes incident to the top 10% inaccurate edges. Colors in the lower method panels indicate the change in absolute restoration error from NI (blue: improved; red: degraded).

NRMSE of the proposed method again increases more gradually than those of the comparison methods across the three graph-construction settings. The proposed method achieves the lowest NRMSE at the larger tested fractions in every setting. These results show that the benefit of magnitude-adaptive regularization is retained even when the two failure modes occur simultaneously. The proposed method does not restore graph information removed by missing edges or weakened by underestimated weights. Nevertheless, limiting excessive penalties due to spurious edges and overestimated weights remains beneficial when such information-loss effects coexist.

It is also worth noting that GTV and GLR+GTV can achieve lower NRMSE than GLR and the proposed method when the proxy graph is accurate or the introduced inaccuracies are mild. This behavior should be distinguished from robustness to proxy-graph inaccuracies because, as discussed in Section III-C, GTV promotes sparse graph differences through its ℓ_1-type evaluation and therefore imposes a signal prior different from the quadratic graph smoothness targeted by GLR and the proposed regularizer. As the excessive-penalty and mixed inaccuracies become more severe, GTV and GLR+GTV exhibit greater performance degradation than the proposed method. In particular, the mixed-inaccuracy experiment shows that the relative performance ordering can change even when the affected-edge fraction for each inaccuracy type is below 10%. These results illustrate the practical importance of limiting sensitivity to excessive penalties when the graph used for recovery is only an imperfect proxy for the underlying graph-smoothness structure.

Figure 1 shows the NRMSE aggregated over the tested inaccuracy levels as a function of τ for the reference graph setting. Across the three inaccuracy experiments, the proposed method exhibits a qualitatively similar dependence on τ, achieving lower NRMSEs in the smaller-τ range and approaching the performance of GLR as τ increases. This consistent behavior complements the quantitative results in Table II by showing the sensitivity of the recovery performance to τ rather than only the performance at the selected common value.

Figure 2 further shows representative recovery results under spurious-edge and overestimated-weight inaccuracies. In both

examples, the proposed method yields the lowest NRMSE among the compared methods and reduces the restoration errors around nodes affected by the introduced inaccuracies.

C. Experiments on Real-World Data

We conduct real-world experiments using meteorological observations from the Global Historical Climatology Network Daily (GHCN-Daily) dataset [50] provided by the National Oceanic and Atmospheric Administration (NOAA). Meteorological observations collected at spatially distributed stations constitute a representative application of graph signal processing, where a graph can naturally be constructed from geographical relationships among the stations. In practice, however, determining graph-construction parameters that appropriately represent the variation of a target meteorological signal is nontrivial, and a graph constructed solely from geographical proximity does not necessarily capture all factors governing the signal variation. Moreover, the appropriate spatial relationships can vary across meteorological variables and daily signals. Consequently, mismatch between the constructed spatial proxy graph and the underlying signal structure can naturally arise even without artificially perturbing the graph. We use this setting to evaluate the practical effectiveness of the proposed regularizer when graph signal recovery relies on such naturally imperfect spatial proxy graphs.

1) Experimental Settings: We consider daily minimum temperature and station pressure measurements recorded at weather stations in the United States during 2017. These two meteorological variables are selected as representative spatial signals whose variation cannot necessarily be explained solely by horizontal geographical proximity. In particular, station pressure can be affected by station elevation as well as horizontal location, making a graph constructed only from inter-station distance a natural example of a potentially imperfect proxy for its variation structure. For each meteorological variable, the measurements on each day are treated as an underlying graph signal. Stations without an available measurement for the corresponding variable and day are excluded before graph construction, and 256 stations are randomly selected from the remaining candidates for each daily signal. The only exception is the station-pressure signal on February 28, 2017, for which only 214 stations with valid measurements are available, resulting in a 214-node graph. Because the set of available stations can vary across variables and days, the node set and spatial proxy graph are constructed independently for each daily signal. The same graph-construction procedure is then applied consistently to all daily signals.

For each daily signal, we construct the spatial proxy graph from the geographic locations of the selected stations using a k-nearest-neighbor (k-NN) rule, in which each station is connected to its k geographically nearest stations. We consider $k \in \{4, 8\}$ to examine different levels of graph connectivity. For each connected station pair (i, j), the edge weight is assigned using the Gaussian kernel $w_{i,j} = \exp(-d_{i,j}^2/(2(\eta\bar{d})^2))$, where $d_{i,j}$ denotes the geographic distance between the two stations and $\eta \in \{1, 2\}$ is the kernel-scale parameter. The reference distance $\bar{d}$ is defined as the median of all distances from each station to its first through eighth nearest neighbors. The four combinations of k and η are evaluated as different plausible constructions of the spatial proxy graph, without assuming that any particular construction exactly represents the underlying graph-smoothness structure.

The observation, graph-weight normalization, optimization, and evaluation settings are otherwise the same as those described in Section V-A. Following the common parameter-selection protocol in Section V-A, for each meteorological variable, the aggregation used to select τ is performed over all daily signals, all four graph-construction settings, and all observation trials.

2) Results: Table III summarizes the annual-average NRMSEs for the daily minimum-temperature and station-pressure signals under the four spatial proxy-graph constructions. For each graph-construction setting, the NRMSE is averaged over all daily signals in 2017 and all observation trials.

For the minimum-temperature signals, GLR, GLR+GTV, and the proposed method achieve very similar annual-average NRMSEs across all four graph constructions, whereas GTV yields consistently higher NRMSEs. The proposed method differs from GLR by at most 0.0002 in these settings and therefore essentially preserves the recovery performance of GLR for the minimum-temperature signals.

For the station-pressure signals, the proposed method achieves the lowest annual-average NRMSE among all compared methods for all four combinations of k and η. In particular, its NRMSE remains lower than those of GLR, GTV, and GLR+GTV regardless of the neighborhood size or Gaussian-kernel scale. These results indicate that the improvement of the proposed method is not restricted to a particular construction of the spatial proxy graph.

Unlike the synthetic experiments, these real-world experiments do not provide an underlying correct graph or labels specifying which graph relations are inaccurate. We therefore do not attribute the observed performance differences to a particular type of proxy-graph inaccuracy. Instead, the results demonstrate the recovery behavior of the compared regularizers when spatial proxy graphs are naturally constructed from geographical relationships under several plausible graph-construction settings. In particular, station pressure provides a relevant example in which geographical distance alone may not fully represent the signal variation because factors such as station elevation are not explicitly incorporated into the graph. The consistent performance of the proposed method across the different spatial graph constructions therefore complements the controlled synthetic experiments and supports its practical effectiveness when recovery relies on naturally imperfect proxy graphs.

To complement the annual-average NRMSE evaluation, Fig. 3 presents a qualitative comparison for a representative daily station-pressure signal. The figure shows the underlying signal, the incomplete noisy observations, and the signals recovered by GLR, GTV, GLR+GTV, and the proposed method using the same spatial proxy graph and a common color scale. In this example, the proposed method achieves the lowest NRMSE of 0.1772 among the compared methods, and the lower panels show smaller restoration errors at several of the

TABLE III

ANNUAL-AVERAGE NRMSES FOR MINIMUM-TEMPERATURE AND STATION-PRESSURE SIGNALS UNDER FOUR SPATIAL PROXY-GRAPH CONSTRUCTIONS (k, η). VALUES IN PARENTHESES ARE SIGNED DIFFERENCES FROM GLR. THE BEST AND SECOND-BEST NRMSES IN EACH COLUMN ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

	Minimum temperature				Station pressure			
Method	$(4,1)$	$(4,2)$	$(8,1)$	$(8,2)$	$(4,1)$	$(4,2)$	$(8,1)$	$(8,2)$
GLR	**0.2643**	<u>0.2610</u>	<u>0.2547</u>	<u>0.2504</u>	0.2019	0.2000	0.1953	0.1926
GTV	0.2758 (+0.0115)	0.2701 (+0.0090)	0.2614 (+0.0067)	0.2539 (+0.0036)	0.2078 (+0.0059)	0.2036 (+0.0035)	0.1972 (+0.0019)	0.1921 (−0.0004)
GLR+GTV	0.2646 (+0.0003)	**0.2609** (−0.0002)	**0.2545** (−0.0002)	**0.2498** (−0.0005)	<u>0.2013</u> (−0.0006)	<u>0.1987</u> (−0.0013)	<u>0.1938</u> (−0.0015)	<u>0.1904</u> (−0.0021)
Proposed	<u>0.2645</u> (+0.0002)	<u>0.2610</u> (0.0000)	0.2549 (+0.0002)	<u>0.2504</u> (0.0000)	**0.1990** (−0.0029)	**0.1974** (−0.0026)	**0.1913** (−0.0040)	**0.1890** (−0.0036)

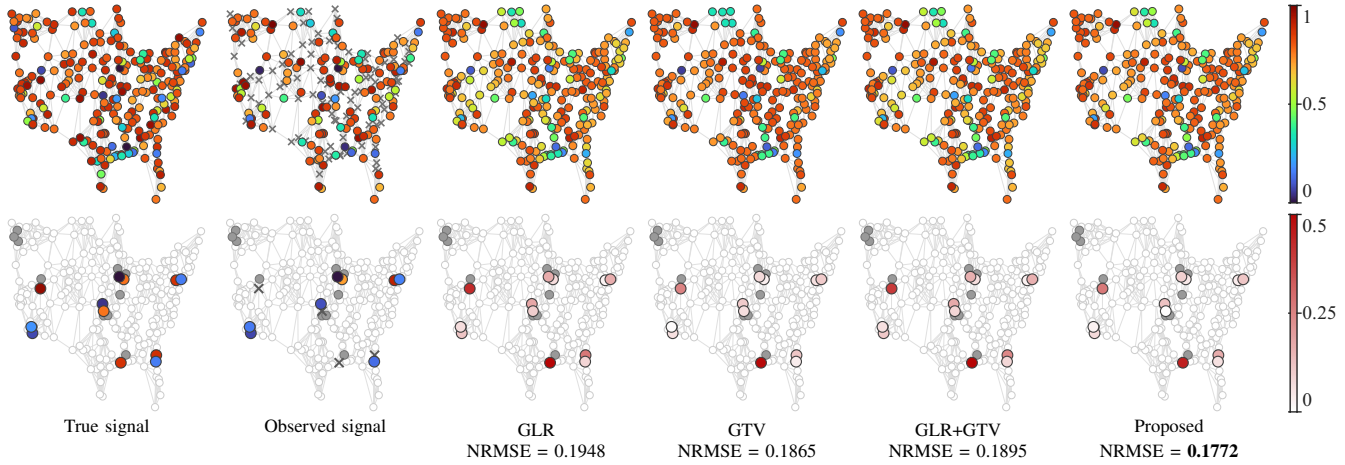


Fig. 3. Qualitative station-pressure recovery for a representative daily signal. The upper row shows the reference, observed, and recovered signals, while the lower row shows the absolute restoration errors at the selected nodes. Gray nodes and edges indicate the surrounding proxy-graph structure.

selected nodes. This qualitative comparison illustrates how the differences among the regularizers appear in the recovered spatial signal and complements the aggregate results reported in Table III.

VI. CONCLUSION

In this paper, we characterized proxy-graph inaccuracies in graph signal recovery through two failure modes, excessive penalties and information loss, and proposed a magnitude-adaptive regularizer derived via infimal convolution and equivalent to a scaled separable Huber penalty to mitigate the former. The proposed regularizer retains GLR-like quadratic evaluation for ordinary proxy-weighted edge differences while limiting the influence of excessively large ones through linear growth, and the resulting constrained convex problem can be efficiently solved by the proposed P-PDS-based algorithm. Synthetic and real-world experiments supported the effectiveness of the proposed method under controlled and naturally imperfect proxy graphs. Developing regularization that can additionally recover graph information lost through missing edges or underestimated weights remains an important direction for future work.

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